LOOKUP FUNCTIONS IN EXCEL

1. What Is a Lookup Function?

A lookup function helps us **find a particular value in a table and retrieve its related information**.

Example: If we know an employee's ID, we can use a lookup function to find their name or salary.

Why do we use lookup functions?

- To find information quickly.
- To retrieve related data from a table.
- To avoid searching through large datasets manually.

2. Types of Lookup Functions

The commonly used lookup functions in Excel are:

1. VLOOKUP
2. HLOOKUP
3. XLOOKUP

3. VLOOKUP (Vertical Lookup)

VLOOKUP searches for a value **vertically in the first column of a selected range** and returns related information from a column to its right.

Syntax

=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])

Arguments

- **lookup_value:** The value we want to find.
- **table_array:** The range of cells containing the data.
- **col_index_num:** The column number from which we want to return the result. The first column of the selected range is column 1.
- **range_lookup:** Specifies the type of match.
    - FALSE – Exact match.
    - TRUE – Approximate match.

Example

Consider the following table:

A (Employee ID)	B (Name)	C (Salary)
101	Rahul	30000
102	Priya	35000
103	Amit	40000

Task: Find the salary of employee 102.

Formula:

=VLOOKUP(102,A2:C4,3,FALSE)

Result: 35000

Explanation:

- 102 is the value we want to find.
- A2:C4 is the table containing the data.
- 3 means return the value from the third column of the selected range.
- FALSE means find an exact match.

Important Points About VLOOKUP

- VLOOKUP searches from **left to right**.
- It searches for the lookup value in the first column of the selected range.
- It returns a value from a column to the right of the search column.
- The column number is counted from the selected range, not from the worksheet.
- If an exact match is not found when using FALSE, Excel returns #N/A.
- VLOOKUP cannot directly return a value from a column to the left of the search column.

4. HLOOKUP (Horizontal Lookup)

HLOOKUP searches for a value **horizontally in the first row of a selected range** and returns related information from a row below it.

Syntax

=HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup])

Example

A	B	C

1 101 102 103

2 Rahul Priya Amit

3 30000 35000 40000

Task: Find the name of employee 102.

Formula:

=HLOOKUP(102,A1:C3,2,FALSE)

Result: Priya

Explanation:

- 102 is the value to find.
- A1:C3 is the selected table.
- 2 means return the value from the second row of the selected range.
- FALSE means exact match.

Important Points About HLOOKUP

- HLOOKUP searches from **top to bottom** within the selected range.
- It searches in the first row of the selected range.
- It returns information from a row below the search row.

5. XLOOKUP

XLOOKUP searches for a value in one range and returns the corresponding value from another range.

Syntax

=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found])

Example

Using the employee table above, find the name of employee 102.

Formula:

=XLOOKUP(102,A2:A4,B2:B4,"Not Found")

Result: Priya

Explanation:

- 102 is the value to find.

- A2:A4 is the range where Excel searches.
- B2:B4 is the range from which Excel returns the result.
- "Not Found" is the message displayed if the value is not found.

Important Points About XLOOKUP

- It can search vertically or horizontally.
- It can return values from either the left or right of the search range.
- It uses an exact match by default.
- It is available in newer versions of Excel, including Microsoft 365 and Excel 2021 or later.

7. Difference Between Lookup Functions

Function	Searches In	Returns From	Search Direction
VLOOKUP	First column	A column to the right	Left to right
HLOOKUP	First row	A row below	Top to bottom
XLOOKUP	Specified range	Specified return range	Left, right, up or down
LOOKUP	One row or column	Corresponding row or column	Depends on the selected vectors

Memory trick:

- VLOOKUP → Vertical search, return to the right.
- HLOOKUP → Horizontal search, return below.
- XLOOKUP → Flexible search in either direction.

FILTER Function

The FILTER function returns all values or rows that satisfy the given condition.

Syntax:

=FILTER(array,include,[if_empty])

Example: Find all employees whose salary is 1400000.

=FILTER(A2:D13,C2:C13=1400000,"No match found")

This formula returns **all matching rows**, including multiple rows with the same salary.

FILTER with Multiple Conditions

We can use multiple conditions to filter data.

Example: Find employees whose job title is Data Analyst and salary is 1400000.

=FILTER(A2:D13,(A2:A13=H8)*(C2:C13=G2),"No match found")

- A2:D13 — Data to return.
- A2:A13=H8 — Checks the job title.
- C2:C13=G2 — Checks the salary.
- * — Represents AND, meaning both conditions must be satisfied.
- "No match found" — Message displayed if no rows match.

Important: If two rows satisfy both conditions, FILTER returns both rows.

Lookup Functions: Behaviour with Duplicate Values

When the lookup value appears more than once, these functions behave as follows:

Function	**Default Behaviour with Duplicates**
VLOOKUP	Returns the first match
HLOOKUP	Returns the first match
XLOOKUP	Returns the first match
XLOOKUP with search mode -1	Returns the last match

Remember: Use FILTER when you want to return **all matching values**, rather than just the first or last match.